Soroosh Shahtalebi | CV

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Research Interests

- Signal Processing
- Machine Learning
- Deep Learning

Education

CIISE Department, Concordia University — **Montreal, QC, Canada**
PhD in Information Systems Engineering, GPA – 4.1/4.3 — *2016–present*
Supervisor: Dr. Arash Mohammadi
Project: Advanced machine learning techniques in bio-signal processing

K.N.Toosi University of technology — **Tehran, Iran**
*M.Sc. in Electrical Engineering (minor in Biomedical) , GPA – 17.86/20 (**1st rank**)* — *2013–2015*
Supervisor: Dr. Reza Jafari
Project: Design and implementation of the processing and controlling circuit board of a blood cell counting machine

Yazd University — **Yazd, Iran**
Bachelors in Electrical Engineering, major in Telecommunication, GPA – 15.28/20 — *2009–2013*
Supervisor: Dr. Jamshid Abouei
Project: Design and production of a planar goniometer for joint angle measurement

Publications

2018: **S. Shahtalebi**, S.F. Atashzar, R.V. Patel, and A. Mohammadi, *SFP-DBRNN: A real-time Signal Filtering and Predicting Framework based on Deep Bidirectional Recurrent Neural Networks*, **(Under review in IEEE Signal Processing Letters)**.

2018: **S. Shahtalebi** and A. Mohammadi, *'A Bayesian Framework to Optimize Double Band Spectra-spatial Filters for Motor Imagery Classification*, IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*(Peer Reviewed)***(accepted)**

2018: **S. Shahtalebi** and A. Mohammadi, *Bayesian Optimized Spatio-Spectral Filters Coupled with Ternary ECOC Classifiers for Motor Imagery BCIs* (**Under review in IEEE Transactions on Neural Systems & Rehabilitation Engineering**).

2018: **S. Shahtalebi** and A. Mohammadi, *Common Spatial Patterns Based on Wavelet Decompositions for Motor Imagery BCIs*, **targeted venue:** IEEE Transactions on Biomedical Engineering, (**to be submitted**).

2017: **S. Shahtalebi** and A. Mohammadi, S.F. Atashzar, and R.V. Patel, *WAKE: Wavelet Decomposition Coupled with Adaptive Kalman Filtering for Pathological Tremor Extraction*, **(Under review in Elsevier journal of Biomedical Signal Processing and Control)(available on Arxiv)**.

2017: **S. Shahtalebi**, A. Mohammadi, S.F. Atashzar, and R.V. Patel, *A Multi-rate and Auto-Adjustable Wavelet Decomposition Framework for Pathological Hand Tremor Extraction*, IEEE Global Signal Processing Conference (GlobalSip), Montreal, Canada *(Peer Reviewed)*.

2017: **S. Shahtalebi** and A. Mohammadi, *Ternary ECOC Classifiers Coupled with Optimized Spatio-Spectral Patterns for Multiclass Motor Imagery Classification*, IEEE International Conference on Systems, Man, and Cybernetics (SMC), Banff, Canada *(Peer Reviewed)*.

2017: **S. Shahtalebi** and A. Mohammadi, *Error Correction Output Codding Coupled with the CSP for Motor Imagery BCI*, 25th European Signal Processing Conference (EUSIPCO), Greece *(Peer Reviewed)*.

2015: **S. Shahtalebi** and R. Jafari, *Design and Implementation of an Electronic Simulator for Aperture-Impedance Electronic Blood Cell Counters*, The 3rd International conference on applied research in electrical, Mechanical and Mechatronic engineering, Tehran, Iran *(Peer Reviewed)*.

2015: **S. Shahtalebi**, J. Abouei, *A New Method to Enhance Accuracy of Goniometry in Joint Angle Measurement*, The 3rd International conference on applied research in electrical, Mechanical and Mechatronic engineering, Tehran, Iran *(Peer Reviewed)*.

Honors and Awards

2018: Concordia Merit Scholarship ($10000), Concordia University, Canada

2018: Selected application for the Deep Learning and Reinforcement Learning summer school, CIFAR and Vector Institute, Toronto, ON, Canada.

2018: Invited visiting researcher to the University of Western Ontario, London, ON, Canada

2018: Concordia Graduate Student Mobility Award ($3000), Concordia University, Canada

2016: Tuition Award of Excellence for International Students ($36000), Concordia University, Canada

2016: Rank 6th (top 1%) in Nationwide Entrance Exam for PhD, National Organization od Educational Testing (NOET), Iran.

2015: Rank 1st in M.Sc. program of Biomedical Engineering (2013 entry), K.N.Toosi University of Technology, Tehran, Iran.

2012: Iran-Patent (NO:77226), *"A Novel Instrument for Sniper Rifle Calibration"*, Iran.

2012: Iran-Patent (NO:76443), *"A New Planar Goniometer for Body Joint Angle Measurement"*, Iran.

2009: Iran-Patent (NO:49444), *"An Automatic and Portable Threading Machine"*, Iran.

2008: Distinguished paper in International Mathematics Competition, *Tournament of Towns*, Moscow, Russia.

2008: Distinguished Idea in Isfahan province, Kharazmi Festival, Iran.

2008: Rank 1st in IranOpen Robotics Competitions, Junior Demo league, Iran.

2008: Winner of the prize *"Most Applied Idea"* in IranOpen Robotics Competitions, Junior Demo league, Iran.

2008: Winner of the prize *"Most Technical Design"* in IranOpen Robotics Competitions, Junior Demo league, Iran.

2007: Nominated Team from Iran for the International Robotics Competitions, Atlanta, USA.

2007: Rank 1st in IranOpen Robotics Competitions, Junior Rescue league, Iran.

Experience

Academic

University of Western Ontario — **London, ON, Canada**
Graduate Visiting Researcher — *2018 (March–Aug)*
Movement Disorders Centre (MDC), London Health Sciences Centre
Under the supervision of Dr. Mandar S. Jog

University of Western Ontario — **London, ON, Canada**
Graduate Visiting Researcher — *2018 (March–Aug)*
Canadian Surgical Technologies and Advanced Robotics (CSTAR)
Under the supervision of Professor Rajni V. Patel

Concordia University — **Montreal**
Lab Demonstrator and Teaching Assistant — *2017 (Sep–Dec)*
Electronics II and Statistics

CIISE Dept., Concordia University — **Montreal, QC, Canada**
Graduate Research Assistant — *2016–present*
Intelligent Signal and Information Processing (I-SIP) Lab
Under Supervision of Dr. Arash Mohammadi

Islamic Azad University (Najafabad Branch) — **Isfahan, Iran**
Lab Director and Instructor, Biomedical Instruments Lab — *2016 (Jan–Aug)*

Institute for Research in Fundamental Sciences (IPM) — **Tehran, Iran**
Research Assistant, School of Cognitive Sciences — *2015 (Sep–Dec)*

K. N. Toosi University of Technology — **Tehran, Iran**
Teaching Assistant, Course: Medical Imaging Systems — *2015*
Under Supervision of Prof. Abrishami Moghaddam

K. N. Toosi University of Technology — **Tehran, Iran**
Teaching Assistant, Course: Statistical Approaches in Biomedical Signal Processing — *2015*
Under Supervision of Dr. Mohebbi

K. N. Toosi University of Technology — **Tehran, Iran**
Graduate Research Assistant, Biomedical Measurements Lab — *2013–2015*
Under Supervision of Dr. Reza Jafari

Industrial

Ubico — **Montreal**
Part-time Technical Consultor — *2017–present*
Providing consultations on how to deploy machine learning techniques for the products and the devices that are ordered by the constumers.

Hit the Books — **Montreal**
Personal Tutor — *2016–present*
Tutoring the undergraduate and graduate courses

Hooshmand Afzar Pars — **Iran**
CEO and Founder — *2015–2017*
The aim of this company was to design biomedical instruments. For the first step, we focused on designing and production of Biofeedback instrumentations and softwares.

Persia Mehr Alvand — **Iran**
Cheif Operating Officer — *2014–2015*
In this position, I had to teach new employees how to do quality certification tests. We were performing biomedical instruments qualification tests in hospitals.

Akhtar Bargh Incs. — **Iran**
Internship — *2012 Jul–Sep*
This company provdies fascilities and infrastructutres for power plants and dispatching stations. As an intern, I was involved in diagnosis and repairment of the telecommunication devices. I was also involved in field studies to determine the communication scheme for the dispatching stations.

Emam Sadegh Institutional and Cultural Foundation — **Isfahan, Iran**
Head of Robotics and Innovations Department — *2009–2011*

In this position, I had to manage courses and strategies to teach robotics to a wide range of students aging from 8 to 18

Computer Skills

Intermediate: Python (tensorflow), LabView, C Programming, PsPice ORCAD, Proteus

Advanced: MATLAB, AutoCAD, Codevision and Assembly for AVR

Relevant Courses

Machine Learning (by Andrew Ng) (*on-line course on Coursera*)

Complete Guide to Tensorflow for Deep Learning with Python (*on-line course on Udemy*)

Machine Learning A-Z; Hands-on Python and R in Data Science (*on-line course on Udemy*)

Probability and Stochastic Processes

Digital Signal Processing

Statistical Pattern Recognition

Biological Signal Processing

Statistical Methods in Signal Processing

Volunteers and Services

2018: Reviewer of the IEEE International Conference on System, Man and Cybernetics (SMC)

2017: Volunteer Organizer for the IEEE GlobalSIP Conference, Montreal, QC, Canada

2017-present: Reviewer of the IET Signal Processing Journal

2016: Volunteer Organizer for the IEEE SPS Winter School, Concordia University, Canada

Language Skills

English: Fluent (TOEFL iBT score:93)

French: Beginner

Persian (Farsi): Mothertongue

References

Dr. Arash Mohammadi: T: (+1) 514-848-2424 Ext. 2712, E: arashmoh@encs.concordia.ca

Assistant Professor,
Concordia Institute for Information System Engineering (CIISE),
Concordia University, Montreal, QC, H3G-2W1, CANADA

Dr. Reza Jafari: T: (+98) 21-8846-2174 Ext. 404, E: jafari@eetd.kntu.ac.ir

Associate Professor
Department of Biomedical Engineering, Faculty of Electrical Engineering
K. N. Toosi University of Technology, Tehran, Iran

Dr. Jamshid Abouei: T: (+98) 353-123-2615, E: abouei@yazd.ac.ir

Associate Professor,
Department of Electrical and Computer Engineering,
Yazd University, Yazd, Iran